INCH-POUND
MIL-PRF-55182G
AMENDMENT 2
12 January 1998
SUPERSEDING
AMENDMENT 1
19 August 1997

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, NON-ESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, AND SPACE LEVEL, GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-PRF-55182G, dated 9 June 1997, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 16 and PAGE 17

TABLE XI, delete and substitute:

"TABLE XI. Qualification inspection.

Inspection	Requirements paragraph	Method paragraph	Number of samples	Number of defects permitted
Certification requirements				
Aqueous-extract-conductivity 1/	3.5.1	4.8		
Flux 1/	3.5.4			2/
Outgassing (space level only)	3.32	4.8.26		
Group I				
Visual and mechanical examination 3/	3.1, 3.4, 3.5, 3.5.1.1 through 3.5.3.2 inclusive, 3.5.5, 3.31, 3.31.1, and 3.32	4.8.1	All sample units	N/A
Thermal shock 4/	3.8	4.8.2		
Overload 4/	3.9	4.8.3		
DC resistance 4/	3.11	4.8.5		
Hermetic seal (when applicable) 4/	3.12	4.8.6		
Group IA				
Solderability	3.13	4.8.7	12 sample units	0
Resistance to solvents	3.14	4.8.8		

See footnotes at end of table.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:35Z
Check the source to verify that this is the current version before use.

TABLE XI. Qualification inspection. - Continued.

Inspection	Requirements paragraph	Method paragraph	Number of samples	Number of defects permitted
Group II				
Resistance temperature characteristics	3.15	4.8.9		
Low temperature storage	3.29	4.8.23	10 highest	
Low temperature operation	3.16	4.8.10	30 10 critical	
Terminal strength	3.17	4.8.11	10 lowest	
Hermetic seal (when applicable)	3.12	4.8.6		
Group III				
Dielectric withstanding voltage	3.18	4.8.12	10 highest	5/ 1
Insulation resistance	3.19	4.8.13	30 10 critical	
Resistance to soldering heat	3.20	4.8.14	10 lowest	
Moisture resistance	3.21	4.8.15		
Group IV				
Shock (specified pulse)	3.22	4.8.16	10 highest	
Vibration, high frequency	3.23	4.8.17	30 10 critical	
Hermetic seal (when applicable)	3.12	4.8.6	10 lowest	
Group V				
Life	3.24	4.8.18	34 highest 102 34 critical 34 lowest	1
Group VA 6/				
+70°C power rating	3.24.3	4.8.18	34 highest 102 34 critical 34 lowest	1
Group VI				
High temperature exposure	3.25	4.8.19	34 highest 7/ 102 34 critical 34 lowest	1
Group VII				
Voltage coefficient (applicable to resistors of 1,000 ohms and over)	3.26	4.8.20		
Fungus	3.27	4.8.21	10 highest	0
Mechanical shear (when applicable)	3.28	4.8.22		

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:35Z
Check the source to verify that this is the current version before use.

TABLE XI. Qualification inspection. - Continued.

Inspection	Requirements paragraph	Method paragraph	Number of samples	Number of defects permitted
Group VIII (space level only) Thermal shock (100 cycles)	3.8	4.8.2	30 (10 highest, 10 critical, 10 lowest)	0

1/ The manufacturer shall verify by certification that these requirements have been met in fabricating resistors furnished to this specification.
2/ Failure of a resistor in one or more tests of a group shall be charged as a single defect.
3/ Marking shall be considered defective if the marking is illegible or incorrect. The two additional unenclosed sample units shall be subjected to the visual and mechanical examination of group I only. Marking shall remain legible at the end of all tests.
4/ As an option, the manufacturer may perform group A, subgroup 1 tests with the required PDA in lieu of these tests.
5/ For characteristic C and characteristic E, zero failures are permitted in moisture resistance.
6/ The power rating shall be +70°C. Qualification inspection only.
7/ Ten sample units shall be subjected to dielectric withstanding voltage and insulation resistance following this test."

PAGE 23

* TABLE XV, add the following new sentence to end of NOTE 1: "(NOTE: Not applicable to space level.)"

PAGE 28

* 4.8.6.b, delete and substitute:

"b. Group I qualification inspection and group A inspection: Method 112 of MIL-STD-202, test condition A or the manufacturer's own procedure, provided the hermetic seal test has been approved by the qualifying activity."

Page 29

* 4.8.9.c, first sentence, delete "+1°C" and substitute "±3°C"

PAGE 45

* Add the new paragraph:

"A.3.3 Outgassing (space level only). Manufacturers shall validate to the qualifying activity the outgassing requirements as specified in 3.32.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:35Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5905-1496)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:35Z
Check the source to verify that this is the current version before use.